Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Remote Sensing, GIS, Artificial Intelligence, and Python

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1. Title

Urban Heat Island Mapping using Land Surface Temperature, Built-up Density, and GIS Analysis.

2. Objective

To analyze spatial patterns of urban heat islands (UHI) by deriving Land Surface Temperature (LST) from satellite data and examining its relationship with built-up density, enabling identification of high-temperature zones and urban thermal hotspots.

3. Study Area

Kolkata Metropolitan Area, West Bengal, India

The study has been conducted over the Kolkata Metropolitan Area (KMA) in the state of West Bengal, India. Kolkata is one of the most densely populated metropolitan regions in the country and has experienced rapid urban expansion over the past few decades. The study area includes major residential, commercial, industrial, vegetated and water-covered regions, making it well suited for Urban Heat Island (UHI) analysis.

The Area of Interest (AOI) was manually delineated using the Geometry tools available in the Google Earth Engine Code Editor to encompass the urban extent of Kolkata and its surrounding regions. The approximate geographic extent of the selected area lies between 22.45°N–22.75°N latitude and 88.20°E–88.55°E longitude.

Kolkata experiences a tropical wet-and-dry climate, characterised by hot and humid summers, a monsoon season, and mild winters. Surface temperatures during summer are often considerably higher within densely built-up areas than in nearby vegetated regions, making the city suitable for studying Urban Heat Island effects.

An important geographical feature within the study area is the Hooghly River, which flows through Kolkata and influences the local microclimate. Water bodies and urban green spaces generally exhibit comparatively lower land surface temperatures than surrounding built-up areas, providing useful contrast for thermal analysis.

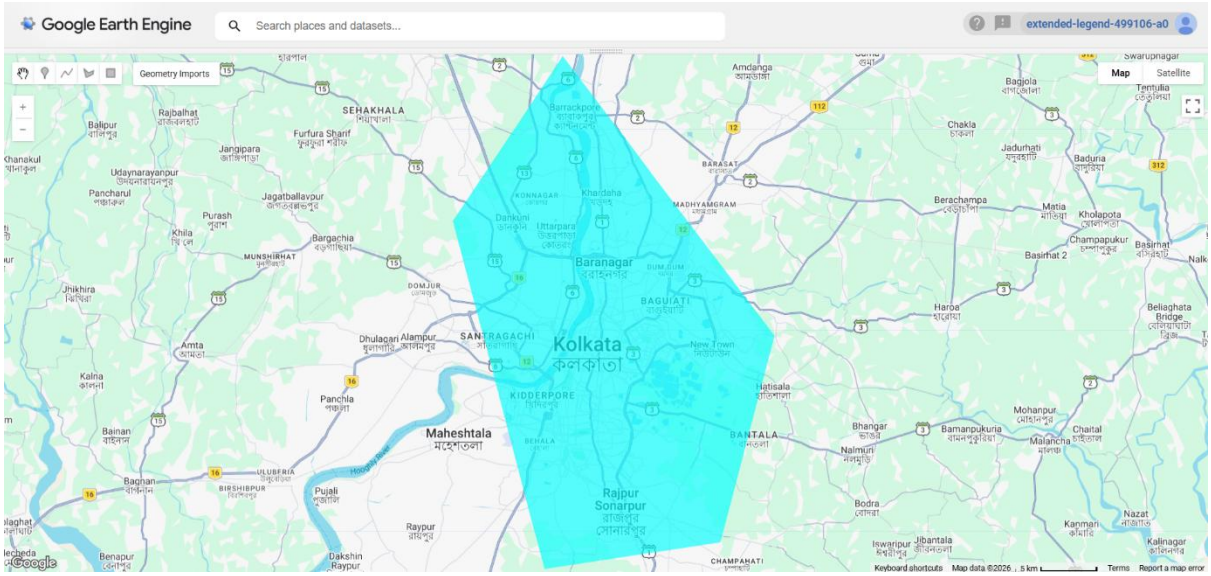


Figure 1. Area of Interest (AOI): Kolkata Metropolitan Area.

4. Data Used

Satellite Data:

• Landsat-8 Collection 2 Level 2

Bands Used:

• ST_B10 (Surface Temperature)

• SR_B4 (Red)

• SR_B5 (Near Infrared)

• SR_B6 (Shortwave Infrared)

Software Used:

• Google Earth Engine

Ancillary Data:

• Area of Interest (AOI) created manually in Google Earth Engine using the Geometry tool.

5. Methodology

Step 1: Area of Interest (AOI) Selection

The Kolkata Metropolitan Area was selected as the study area due to its high level of urbanization and significant variation in land cover and temperature.

Step 2: Landsat-8 Data Acquisition and Preparation

Landsat-8 Collection 2 Level-2 imagery for the year 2023 was obtained from Google Earth Engine. To minimise atmospheric interference and ensure good image quality, only images with less than 10% cloud cover were selected. The Area of Interest (AOI) representing the Kolkata Metropolitan Area was manually delineated using the Geometry tools available in the Google Earth Engine Code Editor. The selected Landsat imagery was clipped to this AOI so that all subsequent analyses were restricted to the study area. A True Colour (RGB) composite was generated using Bands 4 (Red), 3 (Green) and 2 (Blue) to visually inspect the study area before performing thermal and spectral analyses.

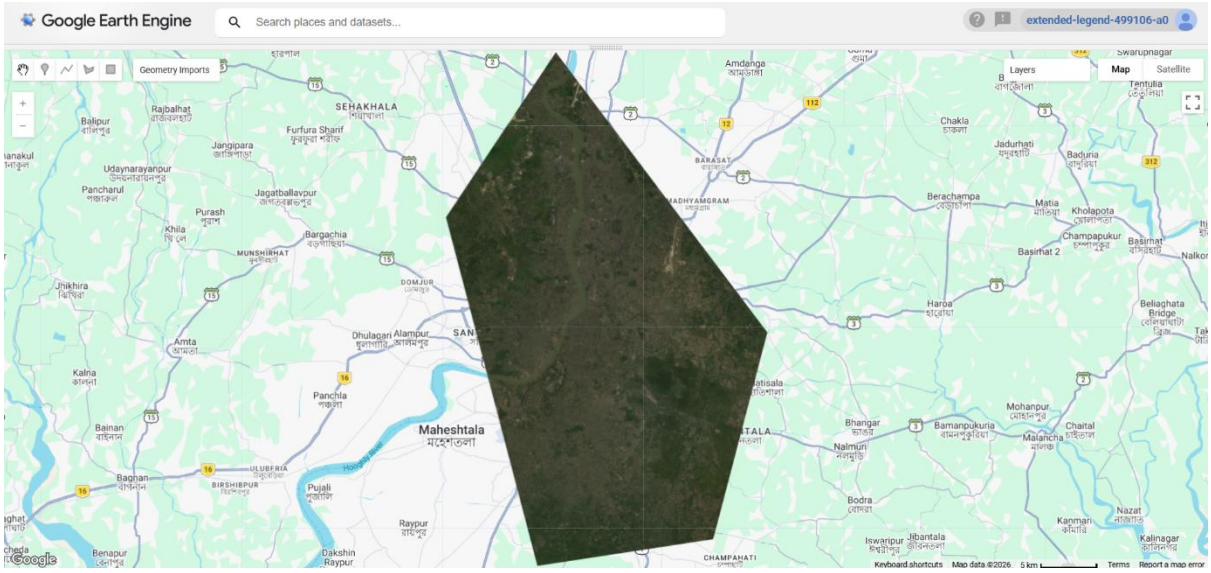


Figure 2. True Color (RGB) composite of the Kolkata study area.

Step 3: Land Surface Temperature (LST) Calculation

The thermal band ST_B10 was extracted and converted into Land Surface Temperature (°C) using the scaling factors provided in the Landsat Collection 2 Level 2 dataset.

Step 4: Normalized Difference Vegetation Index (NDVI) Calculation

NDVI was calculated using the Near Infrared (SR_B5) and Red (SR_B4) bands:
NDVI = (NIR - Red) / (NIR + Red)

Step 5: Normalized Difference Built-up Index (NDBI) Calculation
NDBI was calculated using the SWIR (SR_B6) and NIR (SR_B5) bands:
NDBI = (SWIR - NIR) / (SWIR + NIR)

Step 6: Urban Heat Island (UHI) Identification
Areas having LST greater than 33°C were classified as Urban Heat Island (UHI) hotspots.

Step 7: Hotspot Area Calculation
The area occupied by UHI hotspots was calculated using pixel area analysis in Google Earth Engine and expressed in square kilometres.

Workflow:-

AOI

↓

Landsat-8 Image

↓

LST

↓

NDVI

↓

NDBI

↓

UHI

↓

Area Statistics

6. Results

6.1 Land Surface Temperature

Land Surface Temperature (LST) was derived from Landsat-8 thermal band ST_B10. The temperature distribution over Kolkata ranged from approximately 22°C to 40°C, indicating considerable spatial variation in surface heating.

The generated Land Surface Temperature map reveals considerable spatial variation in surface temperature across the Kolkata Metropolitan Area. Higher temperatures are observed in densely built-up urban regions, while lower temperatures are associated with water bodies and vegetated areas.

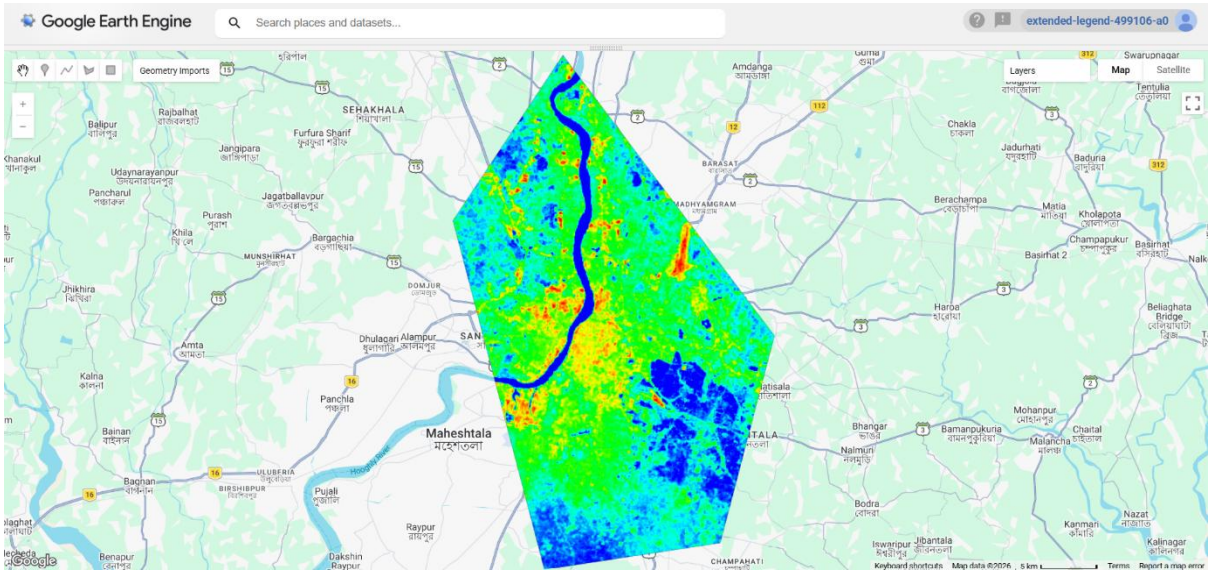


Figure 3. Land Surface Temperature over Kolkata.

Highest temperatures occur in densely urbanized portions of Kolkata, while comparatively lower temperatures are observed near vegetation and river areas.

```
Mean LST                                    JSON
▾Object (1 property)                        JSON
    LST: 28.986625311330716

LST Min Max                                 JSON
▾Object (2 properties)                      JSON
    LST_max: 39.72696890000003
    LST_min: 22.079731639999977
```

Figure 4. Mean, Min & Max LST.

6.2 NDVI Analysis

NDVI was calculated using Landsat-8 Red and Near Infrared bands. Areas with higher NDVI values correspond to vegetation cover, while lower values indicate built-up and barren surfaces.

The NDVI map indicates a predominance of low to moderate vegetation cover across the urbanized Kolkata Metropolitan Area. Higher NDVI values are observed in vegetated and peri-urban regions, while lower values correspond to

densely built-up urban zones. Comparison with the LST map suggests an inverse relationship between vegetation density and surface temperature.

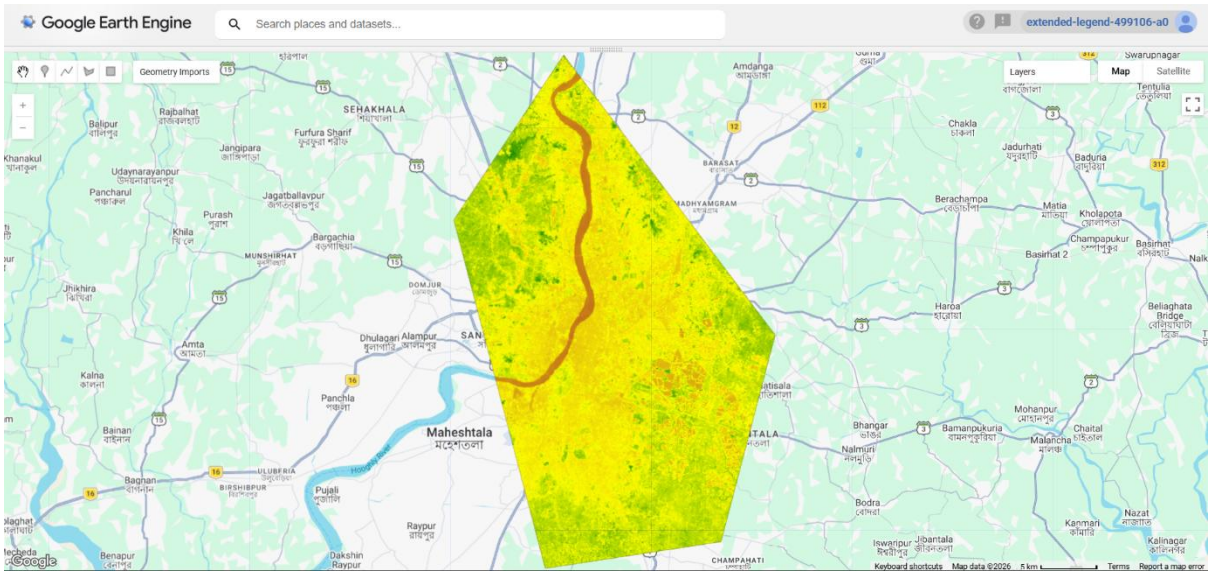


Figure 5. NDVI distribution over Kolkata.

Higher NDVI values indicate healthy vegetation concentrated around peripheral regions. Dense urban areas exhibit low NDVI values.

6.3 Built-up Density Analysis (NDBI)

NDBI was derived using SWIR and NIR bands. Positive NDBI values indicate built-up surfaces, whereas negative values correspond to vegetation and water bodies. The highest built-up concentrations were observed in the urban core of Kolkata.

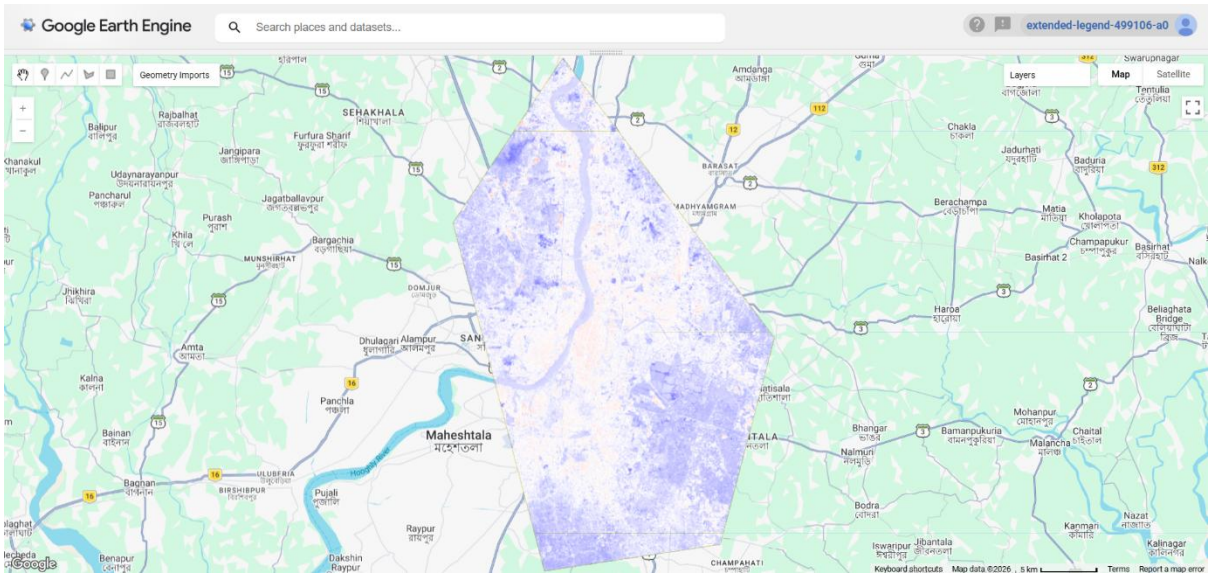


Figure 6. Built-up density (NDBI).

High NDBI values correspond to built-up surfaces and are concentrated in central urban regions.

6.4 Urban Heat Island Zones

The Urban Heat Island (UHI) map was generated by applying a threshold of **33°C** to the Land Surface Temperature (LST) image. Pixels with LST greater than 33°C were classified as UHI hotspots. The analysis revealed a total hotspot area of approximately **19.34 km²** within the selected study area. These hotspots are primarily concentrated in densely built-up urban regions, whereas water bodies and vegetated areas exhibit comparatively lower surface temperatures.

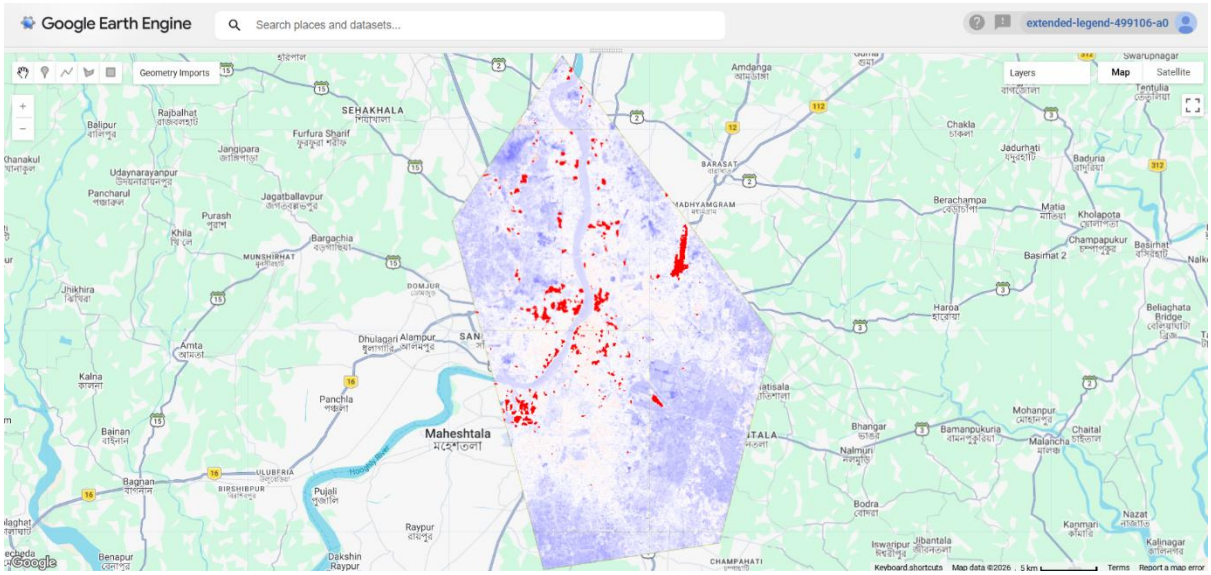


Figure 7. Urban Heat Island hotspots (LST > 33°C).

Urban Heat Island hotspots are concentrated over densely built-up regions having temperatures above 33°C.

```
UHI Hotspot Area (sq. m):                    JSON
▾Object (1 property)                         JSON
   LST: 19342431.46037095

UHI Hotspot Area (sq. km):                   JSON
19.34243146037095
```

Figure 8. UHI Hotspot.

6.5 Built-up Density and Urban Heat Island Relationship

Figure 9 shows the spatial relationship between built-up density (NDBI) and Urban Heat Island (UHI) hotspots. The red hotspot regions are predominantly located within areas having higher NDBI values, indicating that densely built-up regions generally experience higher land surface temperatures. Vegetated and less developed regions exhibit fewer hotspot pixels, demonstrating the cooling effect of vegetation.

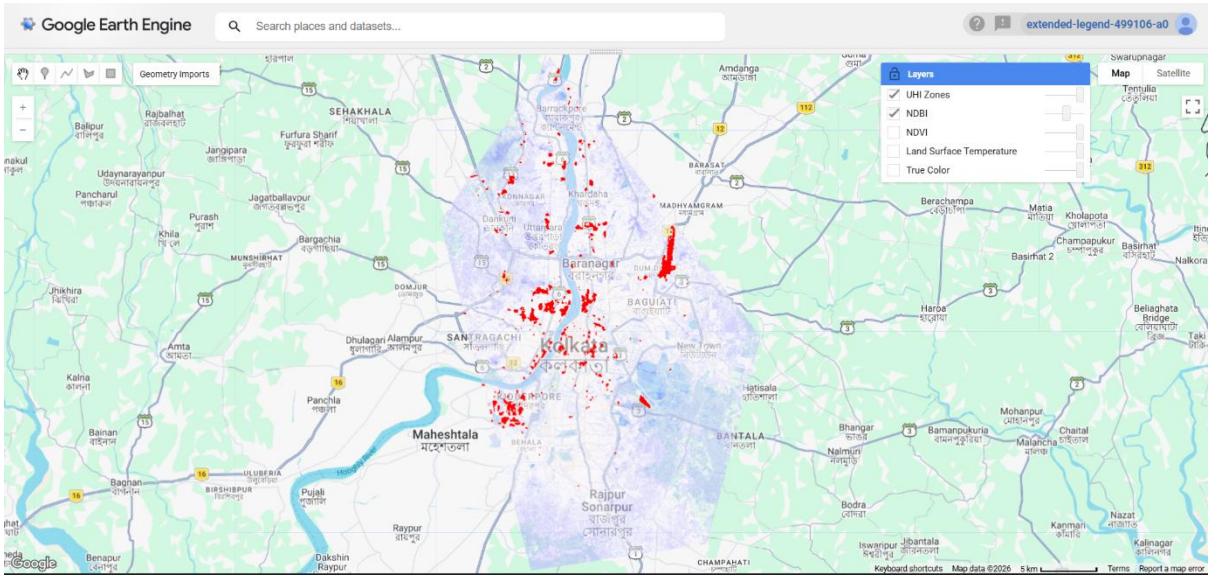


Figure 9. Overlay of Urban Heat Island hotspots on the NDBI map.

Parameter	Value
Mean LST	28.99°C
Minimum LST	22.08°C
Maximum LST	39.73°C
UHI Threshold	33°C
UHI Area	19.34 km²

7. Discussion

- High NDBI → High LST
- Low NDVI → High LST
- Vegetation lowers temperature
- Dense built-up areas coincide with UHI hotspots

The generated Land Surface Temperature map indicates that urbanized regions of Kolkata exhibit comparatively higher temperatures than surrounding vegetated and water-covered areas. The maximum recorded LST was approximately 39.73°C, while the mean temperature was 28.99°C.

The NDVI analysis revealed that areas with higher vegetation cover generally correspond to lower temperatures. This demonstrates the cooling effect of vegetation through shading and evapotranspiration processes.

The NDBI map showed that highly built-up regions are concentrated in the urban core. Comparison of the NDBI and LST maps suggests a positive relationship between built-up density and surface temperature.

The Urban Heat Island map identified approximately 19.34 km² of hotspot area within the study region. Most hotspots were located in densely developed urban areas, while the Hooghly River and vegetated regions exhibited significantly lower temperatures.

Overall, the results indicate that vegetation plays an important role in reducing urban temperatures, whereas extensive built-up development contributes to the Urban Heat Island effect.

8. Limitations

While the adopted methodology provides a reliable assessment of Urban Heat Island patterns over the study area, certain inherent limitations should be considered during interpretation of the results:

- The analysis was conducted using Landsat-8 imagery from a selected period in 2023 and therefore represents conditions during that time only.
- The spatial resolution of Landsat-8 (30 m) may not capture very small urban features and localized temperature variations.
- Urban Heat Island hotspots were identified using a fixed threshold (33°C), which may vary depending on seasonal and climatic conditions.
- The analysis relied primarily on satellite-derived observations and did not incorporate ground-based temperature measurements for validation.

9. Future Scope of the Study

The present analysis can be extended in several ways to provide a more comprehensive understanding of Urban Heat Island dynamics.

- Performing multi-year analysis to evaluate long-term changes in Urban Heat Island intensity.
- Comparing seasonal variations in land surface temperature using imagery from different times of the year.
- Integrating higher-resolution satellite datasets such as Sentinel-2 for more detailed land-cover and built-up analysis.
- Applying machine learning techniques for automated hotspot detection and Urban Heat Island prediction.
- Supporting urban planning by identifying locations suitable for increasing green spaces, improving urban design and implementing heat mitigation strategies.

10. Conclusion

This project successfully analyzed the Urban Heat Island effect in the Kolkata Metropolitan Area using Landsat-8 satellite imagery and Google Earth Engine. Land Surface Temperature, NDVI, and NDBI were generated to study the relationship between temperature, vegetation cover, and built-up density.

The results showed that densely built-up regions generally exhibit higher surface temperatures, while vegetated and water-covered areas remain relatively cooler. Urban Heat Island hotspots covering approximately 19.34 km² were identified using a temperature threshold of 33°C.

The study demonstrates the usefulness of remote sensing and GIS techniques for monitoring urban thermal environments. Such analyses can support urban planning, green infrastructure development, and climate adaptation strategies in rapidly growing cities.

11. References

- Google Earth Engine, https://earthengine.google.com/
- United States Geological Survey (USGS), https://earthexplorer.usgs.gov/
- USGS Landsat 8 Collection 2 Level 2 Dataset

12. Appendix: Google Earth Engine Script

The following Google Earth Engine (GEE) script was used to generate the Land Surface Temperature (LST), NDVI, NDBI and Urban Heat Island (UHI) maps for the study area:

```
var aoi =
    /* color: #00ffff */
    /* shown: false */
    ee.Geometry.Polygon(
        [[[88.34123960929273, 22.814853553694935],
          [88.24510923819898, 22.681874719573557],
          [88.32513843427773, 22.400671122440492],
          [88.47894702802773, 22.422253744984182],
          [88.5263255680668, 22.589088168094047]]]);

Map.centerObject(aoi, 11);

var image = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2')
  .filterBounds(aoi)
  .filterDate('2023-01-01', '2023-12-31')
  .filter(ee.Filter.lt('CLOUD_COVER', 10))
  .median()
  .clip(aoi);

print('Landsat Image', image);

Map.addLayer(
  image,
  {
    bands: ['SR_B4','SR_B3','SR_B2'],
    min: 7000,
```

```
    max: 20000
  },
  'True Color'
);

var lst = image.select('ST_B10')
  .multiply(0.00341802)
  .add(149.0)
  .subtract(273.15)
  .rename('LST');

Map.addLayer(
  lst,
  {
    min: 26,
    max: 36,
    palette: [
      '0000ff',
      '00ffff',
      '00ff00',
      'ffff00',
      'ff8800',
      'ff0000'
    ]
  },
  'Land Surface Temperature'
);

print('Mean LST',
  lst.reduceRegion({
    reducer: ee.Reducer.mean(),
```

```
    geometry: aoi,
    scale: 30,
    maxPixels: 1e13
  })
);
var stats = lst.reduceRegion({
  reducer: ee.Reducer.minMax(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});

print('LST Min Max', stats);

var ndvi = image.normalizedDifference(['SR_B5', 'SR_B4']).rename('NDVI');

Map.addLayer(
  ndvi,
  {
    min: -0.2,
    max: 0.8,
    palette: [
      'brown',
      'yellow',
      'green',
      'darkgreen'
    ]
  },
  'NDVI'
);
```

```
var ndbi = image.normalizedDifference(['SR_B6', 'SR_B5']).rename('NDBI');

Map.addLayer(
  ndbi,
  {
    min: -0.5,
    max: 0.5,
    palette: ['blue', 'white', 'red']
  },
  'NDBI'
);

var uhi = lst.gt(33);

Map.addLayer(
  uhi.updateMask(uhi),
  {palette:['red']},
  'UHI Zones'
);

var hotspotArea = uhi.multiply(ee.Image.pixelArea());
var areaStats = hotspotArea.reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: aoi,
  scale: 30,
  maxPixels: 1e13
});

print('UHI Hotspot Area (sq. m):', areaStats);
print('UHI Hotspot Area (sq. km):', ee.Number(areaStats.get('LST')).divide(1e6)
);
```